

034IN -FONDAMENTI DI AUTOMATICA - FUNDAMENTALS OF AUTOMATIC CONTROL

FULL WRITTEN EXAMINATION - A.Y. 2023/2024

September 16, 2024

PART 1 - Problem 2 - State and Output Response using the Control System Toolbox

Tag List: [STANDARD ASSESSMENT] [PARTIAL CREDIT] [CONTROL SYSTEM TOOLBOX]

Consider the LTI dynamic system described by the following state equations

$$\begin{cases} \dot{x}_1(t) &= -2\,x_1(t) + 16\,u(t) \\ \dot{x}_2(t) &= -8\,x_1(t) - 24\,x_2(t) \\ y(t) &= x_1(t) - 3\,x_2(t) \end{cases}$$

Consider **two** different **scenarios** for the initial state  $x(0)$

$$x_0^{(a)} = \begin{bmatrix} -4 \\ +8 \end{bmatrix} \qquad x_0^{(b)} = \begin{bmatrix} 0 \\ +15 \end{bmatrix}$$

In both cases, the input  $u(t)$  is as follows:

$$u(t) = \left[ 16 + \cos(2.3\,t) - 4.2\,\sin\,(23\,t) \right] \cdot \mathbf{1}(t)$$

As usual,  $\mathbf{1}(t)$  denotes the unit step function:  $\mathbf{1}(\tau) = \begin{cases} 0 & \longleftrightarrow \tau < 0 \\ 1 & \longleftrightarrow \tau \geq 0 \end{cases}$   $\mathbf{1}(\tau) = \begin{cases} 0 & \longleftrightarrow \tau < 0 \\ 1 & \longleftrightarrow \tau \geq 0 \end{cases}$

**Determine** the state response  $x(t)$  and the output response  $y(t)$  according to:

$$x(t) = e^{At}x_0 + \int_0^t e^{A(t-\tau)}Bu(\tau)\,d\tau$$

$$y(t) = Ce^{At}x_0 + \int_0^t Ce^{A(t-\tau)}Bu(\tau)\,d\tau$$

in the time interval

$$t \in [0, 10]$$

Assignment

Your code shall:

- Create the **numeric** matrices **A**, **B**, **C** and **D** according to the system description.
- Create two **numeric** array for the possible initial states **x0a** and **x0b**.
- Create an array with time instants in the given interval [0, 10]. Use the function **linspace( )** and create an array with **50000** time instants, named **t**.
- Using the already defined array t, compute **50000** samples of the input function  $u(t)$ . Assing the result to the **numeric array** named **u**.
- **Scenario (a)**: Compute the state and the output response when the initial state is **x0a** and the input is **u**. Use the function **lsim( )** ONLY. Assign the resulting state and output responses to the variables **x\_a** and **y\_a**.
- **Scenario (b)**: Compute the state and the output response when the initial state is **x0b** and the input is **u**. Use the function **lsim( )** ONLY. Assign the resulting state and output responses to the variables **x\_b** and **y\_b**.

## Solution

Compare with Grader-State & Output Trajectory of ... - Problem 5

```
clear

% system matrices
A = [      -2      0;
      -8     -24];
B = [16 ; 0];
C = [1  -3];
D = 0;

% time array
t = linspace(0, 10, 50000);

% input array
u = (16+cos(2.3*t)-4.2*sin(23*t)).*heaviside(t);

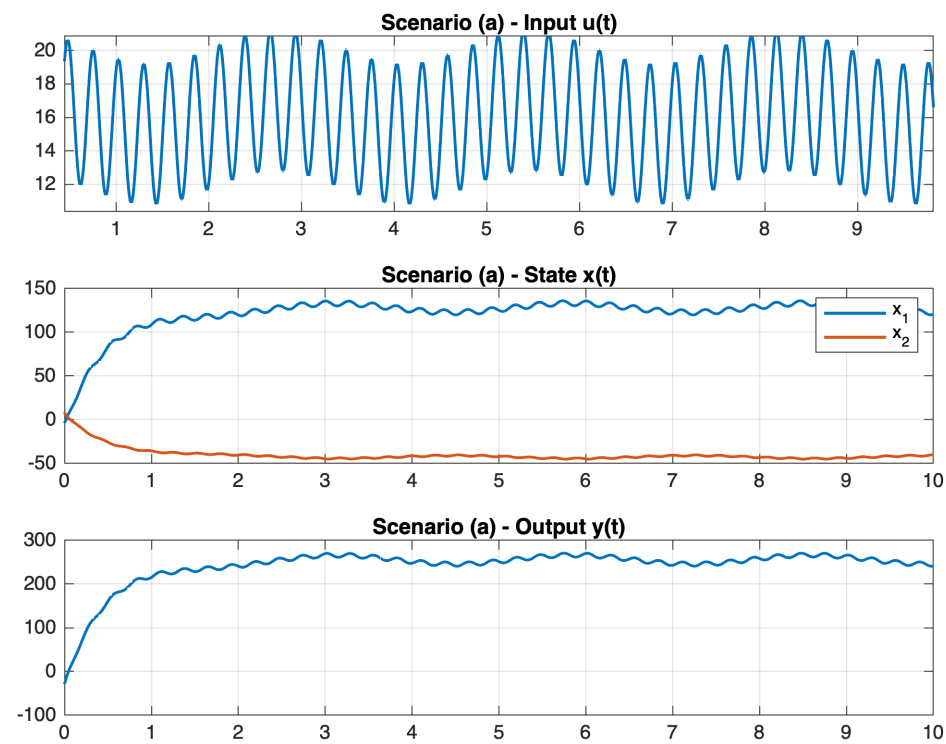
% LTI system
LTISYS = ss(A,B,C,D);

% initial state
x0a = [-4; +8]; x0b = [0; +15];

% scenario (a)
[y_a, ~, x_a] = lsim(LTISYS,u,t, x0a );

% plot the results -- NOT REQUIRED
figure;
subplot(3,1,1);
plot(t,u, 'linewidth', 1.5);
title('Scenario (a) - Input u(t)')
grid on; zoom on
subplot(3,1,2);
plot(t, x_a, "LineWidth", 1.5);
title('Scenario (a) - State x(t)')
legend('x_1', 'x_2')
grid on; zoom on
subplot(3,1,3);
plot(t, y_a, "LineWidth", 1.5);
title('Scenario (a) - Output y(t)')
```

grid on; zoom on



```
% scenario (b)
[y_b, ~, x_b] = lsim(LTISYS,u,t, x0b );

% plot the results
figure;
subplot(3,1,1);
plot(t,u, 'linewidth', 1.5);
title('Scenario (b) - Input u(t)')
grid on; zoom on
subplot(3,1,2);
plot(t, x_b, "LineWidth", 1.5);
title('Scenario (b) - State x(t)')
legend('x_1', 'x_2')
grid on; zoom on
subplot(3,1,3);
plot(t, y_b, "LineWidth", 1.5);
title('Scenario (b) - Output y(t)')
grid on; zoom on
```

